Project Management Plan

ClearanceDNSC

Proposed Web-Based Student Clearance Management System

Document control	Details
Organization	Davao del Norte State College (DNSC)
Project name	ClearanceDNSC
Document type	Project Management Plan with System Flow and Requirements
Version / Date	Draft Version 1.0 | October 28, 2026
Project sponsor / owner	Proposed Institutional Stakeholders: DNSC
Project manager	Elyza M. Flores
Project status	In Progress — Planning and Requirements Definition

1. Purpose and Management Approach

This Project Management Plan describes how ClearanceDNSC will be planned, designed, developed, tested, demonstrated, and handed over. The proposed system is a web-based platform intended to help students initiate and track clearance requests while authorized office personnel verify requirements and record decisions.

This team will use a phased approach: project initiation, requirements gathering, system and interface design, development, testing, user evaluation, deployment or demonstration, and project closure. Changes that affect the agreed scope, schedule, or budget will be recorded and reviewed by the team and instructor before implementation.

1.1 Project Objectives

- **Digitalize the Clearance Process** – Provide a web-based platform where students can initiate clearance requests and authorized officers can process clearance items.

- **Reduce Paperwork and Manual Follow-ups** – Minimize dependence on physical clearance forms and unnecessary office visits.
- **Improve Security and Accountability** – Implement role-based access control and maintain records of important clearance decisions.
- **Enable Clearance Progress Tracking** – Allow students to monitor the status of their clearance items and identify pending requirements.
- **Prepare Users for System Adoption** – Provide user guides, system demonstrations, and basic instructions to help students, clearance officers, and administrators use the platform.

2. Requirements Management Plan

Requirements should be gathered from the project instructor, student representatives, and any available DNSC representatives or staff who understand the clearance process. The team will document each requirement, review it with stakeholders, and link it to design work and test cases.

Control activity	Management approach
Document	Record functional, non-functional, data, privacy, reporting, and usability requirements in the requirements register.
Validate	Review requirements with the instructor and relevant stakeholders before treating them as approved.
Trace	Use the requirements traceability matrix to connect each requirement to a module, test case, and acceptance result.
Control changes	Record proposed changes in a change log and assess effects on scope, schedule, cost, risks, and workload before approval.

2.1 Proposed Functional Requirements

ID	Requirement	Primary user	Acceptance measure
FR-01	The system shall allow authorized users to	All roles	Valid and invalid log in tests

	log in and log out.		pass.
FR-02	The system shall display features according to the user's assigned role.	All roles	Student, officer, and admin access tests pass.
FR-03	Students shall be able to view their profile and own clearance records.	Student	A student cannot access another student's record.
FR-04	Students shall be able to initiate a clearance request when eligible.	Student	A valid request is saved and shown in the dashboard.
FR-05	The system shall show the status of each required office or clearance item.	Student	Status and remarks match the saved record.
FR-06	Officers shall view requests assigned to their office.	Officer	An officer cannot process another office's items.
FR-07	Officers shall be able to approve or reject an assigned item and add remarks when needed.	Officer	Decision, actor, and timestamp are saved.
FR-08	The system shall update overall progress after an authorized decision.	System	Progress reflects all required items.
FR-09	The system shall not mark a clearance completed until all required approvals are obtained.	System	Incomplete clearance cannot become completed.
FR-10	Administrators shall manage user accounts, roles, offices, and clearance requirements.	Admin	Authorized changes save and unauthorized changes are blocked.
FR-11	The system shall maintain a history of important clearance actions.	System / Admin	Action history shows the user, action, and time.
FR-12	The system should provide notifications when clearance status changes.	System / Officer	Notification appears if this feature is included.
FR-13	The system should generate reports for pending, approved, rejected, and completed clearances.	Admin	Report totals match the test data.
FR-14	The system may allow authorized students to print or download a completed clearance.	Student	Output is available only after completion, if included.

2.2 Proposed Non-Functional Requirements

ID	Requirement

NFR-01 Security	Passwords must be stored using secure password hashing and access must be role-based.
NFR-02 Privacy	Student information must be visible only to authorized users for legitimate project purposes.
NFR-03 Usability	Labels, statuses, and actions must be clear to students and office personnel.
NFR-04 Responsiveness	Pages should work on common desktop and mobile screen sizes.
NFR-05 Compatibility	The system should support current versions of common web browsers used for testing.
NFR-06 Reliability	Saved records and decisions should remain consistent after refresh or re-login.
NFR-07 Auditability	Important decisions must record who acted and when.
NFR-08 Backup	The team should document a database backup and recovery procedure for the deployed version.
NFR-09 Maintainability	Code, database tables, and setup instructions should be organized and documented.
NFR-10 Data integrity	The system should prevent duplicate clearance requests for the same student and clearance period, where applicable.

2.3 Sample Requirements Traceability Matrix

ID	Requirement summary	Source / stakeholder	Test / acceptance evidence
REQ-01	Role-based access	Instructor / project team	Role permissions tested for each account type.
REQ-02	Student can track clearance progress	Student representative	Dashboard displays correct office statuses.
REQ-03	Office can record decision and remarks	Clearance-proces s stakeholder	Approval/rejection saves actor, timestamp, and remarks.
REQ-04	Completion requires all required approvals	Instructor / project team	A pending or rejected required item prevents completion.
REQ-05	Protect student information	Instructor / DNSC policy to confirm	Unauthorized access attempts are blocked.
REQ-06	Generate clearance status	Administrator /	Report totals match saved

	report	project team	sample records.

3. Scope Management Plan

3.1 Proposed In-Scope Deliverables

The proposed ClearanceDNSC project shall include the following deliverables, subject to confirmation by the project manager:

- A responsive web interface with login and role-based dashboards.
- Student clearance request and progress-tracking pages, including a checklist of required clearance items.
- Wireframes, prototypes, and interface designs for the student dashboard, clearance officer portal, and administrator portal.
- Office review and approval workflows with status remarks for pending or rejected clearance items.
- Administrator functions for managing user accounts, roles, offices, clearance requirements, and basic reports.
- Backend functions for authentication, routing, clearance processing, status updates, and role-based access control.
- Database structures for users, students, offices, clearances, clearance items, and action history, including appropriate relationships and integrity constraints.
- API specifications and integration functions for approved services or department portal communication. Mock integrations may be used when official access is unavailable.
- QR code generation and verification for the approved master clearance pass or clearance reference.
- Automated email notifications for approved clearance events.

- Security measures for authentication, authorization, session handling, input validation, sensitive information, and database access.
- Database backup and restoration procedures for the selected development or deployment environment.
- Quality assurance documents, including a test plan, test cases, defect log, test results, and retesting records.
- Mobile responsiveness, functional, integration, security, and edge-case testing.
- Cloud deployment of the web application to the hosting environment selected by the project team, together with deployment configuration and setup documentation.
- Basic reports and a printable or downloadable final clearance, if approved for the first version.
- Mock student accounts, department accounts, and simple data for testing and live demonstration.
- Presentation slides, promotional graphics, and an overview video, when required for the project launch.
- A user guide, technical setup notes, project documentation, and final project report.

3.2 Proposed Out-of-Scope Items

- Direct integration with the official DNSC student information system unless access and approval are provided.
- Online payment processing, accounting integration, or automated financial clearance.
- Biometric identification, native mobile applications, and SMS gateway integration.
- Legally binding electronic signatures or digital certificates unless specifically required and approved.

- Automatic verification from external office databases not available to the team.
- Any production use of real student data without institutional authorization.
- Replacement of the existing DNSC Student Information System or development of a complete student information management platform beyond the student clearance workflow.

3.3 Scope Control

- The team lead maintains the scope statement, deliverables, and task breakdown.
- Completed features are checked against acceptance criteria before being marked complete.
- New features or policy changes are entered in the change log and reviewed before work begins.
- The instructor or designated project authority confirms major scope changes.
- Charter Alignment: The project charter shall serve as the high-level reference for the approved project objectives, scope, major deliverables, and target schedule. The Project Management Plan shall provide the detailed implementation approach. Any proposed changes that create inconsistencies between these documents must be recorded in the change log and reviewed by the project manager and project instructor or designated project authority before implementation.

3.4 Work Breakdown Structure (WBS)

WBS	Work package	Key output	Responsible department
1.0	Project Management and Documentation	Project plan, meeting minutes, status updates, change log and requirements records	Executive & Project Management Office
2.0	Requirements and Process Analysis	Validated requirements, clearance workflow, use cases, and business rules	Executive & Project Management Office, with input from all departments

3.0	UI/UX Design	Wireframes, prototypes, navigation flow, and interface specifications	UI/UX & Frontend Development
4.0	Frontend Development	Student dashboard, clearance tracker, officer portal, administrator portal, and responsive pages	UI/UX & Frontend Development
5.0	Backend Development	Authentication, routing, clearance processing, role-based permissions, and status update logic	Backend, Database & API Integration
6.0	Database and Security	Database schema, relationships, integrity constraints, access controls, and backup procedures	Backend, Database & API Integration
7.0	API, QR Code, and Email Integration	API specifications, approved integrations or mock services, QR verification, and email notification workflows where included	Backend, Database & API Integration
8.0	Quality Assurance and System Testing	Test plan, test cases, defect log, test results, integration testing, and retesting records	Quality Assurance & Testing
9.0	Deployment and DevOps	Environment configuration, deployment documentation, and system availability checks	Deployment, DevOps & Presentation
10.0	Presentation and Live Demonstration	Presentation slides, media materials, mock accounts, test data, and demonstration checklist	Deployment, DevOps & Presentation
11.0	Final Review and Handover	Final acceptance review, user guide, technical documentation, and final project report	Executive & Project Management Office, with contributions from all departments

4. Proposed System Flow

The ClearanceDNSC system follows a structured workflow that allows students to submit clearance requests, clearance officers to review and process assigned clearance items, and administrators to manage system operations. The system coordinates the frontend interface,

backend processing, database operations, and approved integrations to ensure that clearance information is recorded and displayed consistently.

4.1 Technical Coordination Across the System Flow

1. **Login and Role Identification:** The system verifies user credentials and determines whether the user is a student, clearance officer, or administrator. Access to system functions is based on the user's assigned role and permissions.
2. **Student Clearance Request:** The student initiates a clearance request and views the required clearance items assigned to the applicable clearance process.
3. **Office Review:** The authorized clearance officer reviews assigned clearance items and records an approval, rejection, or pending status, including remarks when required.
4. **Backend Processing:** The backend validates each action, checks the user's permissions, processes the request, and saves the decision and relevant action history.
5. **Database Update:** The database stores the clearance decision and related information while maintaining valid relationships, data consistency, and integrity.
6. **Progress Tracking:** The student dashboard displays the latest saved clearance status and overall progress, allowing students to monitor which clearance items have been approved or still require action.
7. **API Integration:** API and Integration Specialists support communication between authorized system components or approved external services. If official API access is unavailable, a mock integration may be used for demonstration, subject to the approved project scope.
8. **QR Verification:** The QR Code Engineers shall implement QR code generation, scanning, and verification for the approved digital clearance reference or master

clearance pass. The verification process shall confirm the validity of the clearance reference without exposing sensitive student information or bypassing server-side authorization.

9. **Email Notification:** The Email Automation Engineers shall configure automated email notifications for confirmed clearance events, including notification when a student's clearance is fully completed. The notification process shall be tested to ensure that messages are triggered by the appropriate system events.
10. **Quality Assurance:** The Quality Assurance (QA) team tests the end-to-end process, including user permissions, saved decisions, status updates, integration behavior, responsive layouts, and relevant edge cases. Identified defects are documented, corrected, and retested.
11. **Deployment and Demonstration:** The Deployment, DevOps and Presentation Department shall prepare the agreed cloud hosting environment, configure the application for deployment, and verify that the system is accessible for the planned demonstration. The team shall prepare mock accounts, sample data, presentation materials, and a demonstration checklist. Deployment and demonstration activities shall be completed within the approved project schedule.

4.2 User Roles

1. **Student:** Students are the primary users who initiate and monitor their clearance requests. Their responsibilities and permitted activities include:
    - Logging in to their assigned accounts.
    - Viewing their clearance checklist and required clearance items.
    - Monitoring the approval status and overall progress of their clearance.
    - Viewing relevant remarks or instructions provided by authorized clearance officers.

- Accessing the final clearance reference or printable clearance document if this feature is approved and implemented.

2. **Clearance Officer:** Clearance officers are responsible for reviewing and processing clearance items assigned to their respective offices. Their responsibilities and permitted activities include:
    - Logging in to their authorized accounts.
    - Viewing clearance requests assigned to their respective offices.
    - Reviewing the information required to process each clearance item.
    - Approving or rejecting clearance items and recording pending statuses when applicable.
    - Providing remarks or instructions when required.
    - Updating clearance decisions through the system, subject to role-based permissions.
3. **Administrator:** Administrators oversee the system's configuration, user management, and general operations. Their responsibilities and permitted activities include:
    - Managing user accounts and assigned roles, subject to approved administrative permissions.
    - Maintaining office assignments and required clearance items.
    - Monitoring clearance records and overall system activity.
    - Managing relevant system settings and supporting data maintenance.
    - Assisting with troubleshooting and coordinating system operations.

4.3 Core Business Rules

The following business rules shall guide the operation of the ClearanceDNSC system:

1. **Account Authentication:** Users must authenticate before accessing protected system functions.

2. **Role-Based Access:** Students, clearance officers, and administrators shall have different permissions based on their assigned roles.
3. **Clearance Request:** A student must initiate a clearance request before the applicable clearance items can be processed, following the workflow approved by the project manager.
4. **Office Assignment:** Each clearance item must be associated with the appropriate office or authorized clearance officer.
5. **Clearance Decision:** Only an authorized clearance officer may approve or reject a clearance item assigned to their office.
6. **Status Recording:** Each clearance decision must be saved in the database and reflected in the appropriate system views.
7. **Remarks and Instructions:** Clearance officers may provide remarks or instructions when a clearance item is rejected, remains pending, or requires further action, according to the confirmed workflow.
8. **Overall Clearance Status:** A student's overall clearance status shall become **Completed** only when all required clearance items have been approved. Otherwise, the system shall display the applicable incomplete or in-progress status based on the confirmed business rules.
9. **Data Integrity:** The system must maintain consistent relationships between student accounts, clearance requests, clearance items, office assignments, and recorded decisions.
10. **QR Code Security:** If QR verification is implemented, scanning the QR code must not automatically grant access to protected records or authorize clearance approval. Verification must follow the approved security process.
11. **Email Notifications:** If email automation is implemented, notifications shall be triggered only by the specified events included in the approved project scope.

12. **Data Privacy:** Real student information may only be used with appropriate authorization. Mock or sample data shall be used for development and demonstrations when real data access is not approved.
13. **Integration Limitations:** External APIs and services may only be integrated with the necessary authorization and access. Mock integrations may be used when official access is unavailable and the project scope permits them.
14. **Auditability:** Relevant clearance actions should record sufficient information to support accountability and troubleshooting, subject to the approved data collection and retention requirements.
15. **Final Clearance Document:** A printable or downloadable final clearance document or reference may be provided only if this feature is included in the approved scope and its completion requirements have been defined.

5. Schedule Management Plan

The Schedule Management Plan defines how project activities, deadlines, milestones, and deliverables will be organized and monitored. The project team shall use the Work Breakdown Structure (WBS) to identify tasks, assign responsibilities, establish target completion dates, and monitor progress.

The schedule shall be reviewed regularly by the project manager and team members. Delays, dependencies, and changes affecting major deliverables shall be documented and communicated to the team and project instructor. The schedule may be adjusted based on the approved project timeline, resource availability, and the results of testing and evaluation.

5.1 Milestones

The following milestones represent the proposed development schedule for ClearanceDNSC. The exact dates shall be finalized based on the course requirements and the project instructor’s approved timeline.

Milestone	Key Activities	Expected Deliverables	Target Schedule
M1: Project Initiation and Planning	Confirm objectives, stakeholders, team assignments, requirements, and clearance workflow. Begin database planning and authentication setup.	Reviewed project plan, initial requirements, and proposed system flow	Week 1
M2: Interface and Core Development	Develop wireframes and interfaces for the student dashboard, clearance officer portal, and administrator portal. Begin backend and database integration	Reviewed interface designs and functional core modules	Week 2
M3: System Integration and Feature Completion	Integrate frontend, backend, and database. Implement QR code generation and verification, email notifications, and cloud deployment preparation.	Integrated system prototype with confirmed features	Week 3
M4: Testing and Corrections	Conduct functional, integration, security, usability, and responsive-layout testing. Record defects and retest corrected features.	Test results, defect log, and corrected system	Week 4
M5: Deployment, Demonstration and Handover	Deploy the application to the agreed cloud environment, prepare presentation materials, demonstrate the system, and submit documentation.	Functional web application, database schema, QR verification system, documentation, and deployment acceptance report.	Week 4

5.2 Resource Requirements

The successful implementation of ClearanceDNSC requires human, technical, software, and documentation resources. The project manager shall coordinate resource allocation and monitor whether the available resources are sufficient to complete the agreed deliverables.

Resource Category	Required Resources	Purpose

Human Resources	Executive and Project Management Office; UI/UX and Frontend Development; Backend, Database and API Integration; Quality Assurance and Testing; Deployment, DevOps and Presentation	Manage, analyze, design, develop, integrate, test, deploy, document, and present the system
Hardware Resources	Laptops or desktop computers and reliable internet connectivity	Support development, collaboration, system testing and online demonstrations
Software Resources	Code editor or IDE, web browsers, database management software, version control platform, and testing tools	Support programming, database development, collaboration, and quality assurance
Development Technologies	Approved frontend and backend technologies, database system, and web server, QR code generation and scanning tools, and email automation service	Develop the web-based clearance system and implement its required features
Design and Documentation Tools	Wireframing or prototyping software, word processing software, spreadsheets, and presentation tools	Prepare user interface designs, project documents, reports, and presentation materials
Testing Resources	Mock student accounts, clearance officer accounts, administrator accounts, sample clearance records, and test cases	Verify system functionality, role-based access, clearance workflows, QR verification, and email notifications without using unauthorized real student records
Deployment Resources	Local development environment, approved cloud hosting environment, database backup facilities, deployment configuration, and setup documentation	Support cloud deployment, system demonstration, and recovery procedures
Stakeholder Resources	Project instructor, student representatives, and authorized DNSC representative or staff, where available	Validate requirements, provide feedback, and review the proposed workflow

Resource allocation shall be based on team members' skills and availability. Each assigned task shall have a designated owner, expected output, and target completion

date. If a required tool, integration, or hosting service is unavailable, the team may use an approved alternative or mock service, provided that the substitution is documented.

6. Cost Management Plan

6.1 Budget Control

7. Performance Measurement and Acceptance

Project performance shall be measured by monitoring task completion, adherence to the approved four-week schedule, completion of required system features, and compliance with the project charter. Department leads shall report their progress, identify delays or technical issues, and provide evidence of completed work to the Project Manager.

The system shall be evaluated through functional testing, integration testing, security checks, usability testing, and responsiveness testing. Test results and identified defects shall be documented, and necessary corrections shall be completed before the final demonstration and handover.

The project shall be considered ready for acceptance when the required features are implemented and tested, critical defects are resolved, the system is deployed to the agreed cloud hosting environment, and the required documentation and demonstration materials are prepared. Final acceptance shall be subject to review by the Project Manager and the project instructor or designated project authority.

7.1 Minimum Acceptance Tests

The following acceptance tests shall be performed to verify that ClearanceDNSC meets its minimum functional, security, and deployment requirements. Each test shall be documented as passed or failed, with remarks recorded for any issue requiring correlation.

Test ID	Acceptance Test	Expected Result
AT-01	User Authentication	Authorized users can log in and log out using valid credentials. Invalid credentials shall not

		grant access.
AT-02	Role-Based Access Control	Students, clearance officers, and administrators can access only the features and information permitted for their assigned roles.
AT-03	Student Dashboard	Students can view their clearance progress and the status of their required clearance items.
AT-04	Officer Approval Workflow	Clearance officers can view and process clearance items assigned to their respective offices. They can approve or flag requests and provide remarks where applicable.
AT-05	Clearance Status Updates	The system records approval decisions and updates the corresponding clearance status accurately.
AT-06	Overall Clearance Completion	A student's clearance shall be marked as complete only when all required clearance items have been approved.
AT-07	Administrator Functions	Authorized administrators can manage user accounts, roles, offices, and clearance requirements.
AT-08	QR Code Verification	The system can generate a QR code and verify the associated clearance reference through the implemented scanning or verification process. Invalid or altered references shall not be treated as valid clearance.
AT-09	Email Notifications	The system sends an automated email notification when a student's clearance is fully completed, using the configured email service and test accounts.
AT-10	Data Security and Integrity	Unauthorized users cannot access restricted functions, and clearance records remain consistent after authorized updates.
AT-11	Responsiveness and Usability	The main student, officer, and administrator interfaces remain usable on supported desktop and mobile screen sizes.
AT-12	Cloud Deployment	The application is accessible through the agreed cloud hosting environment, and its essential functions work in the deployed version.
AT-13	Documentation and Handover	The team provides the required database schema, system documentation, user or setup

		guide, test results, and deployment acceptance report.

8. Risk and Issue Management

Control activity	Management approach
Identify	Proactively flag potential risks (e.g., technical roadblocks, schedule delays, scope creep) and active issues as they arise during development and testing.
Assess	Evaluate the likelihood and potential impact of each risk or issue on project scope, timeline, and quality.
Mitigate	Formulate and document preventive actions for identified risks and corrective actions for active issues.
Log & Track	Maintain a centralized risk and issue log to monitor status, ownership, and resolution progress through project closure.

8.1 Proposed Risk Register

Risk ID	Potential risk	Impact	Likelihood	Mitigation / Response strategy
RSK-01	Unauthorized users attempt to access or modify other students' clearance records or office approvals.	High	Medium	Implement role-based access control, validate permissions on the server, and test access restrictions for all user roles.
RSK-02	Clearance statuses or database records become inconsistent after updates, page refreshes, or re-login.	High	Medium	Conduct integration testing, validate database relationships, and verify that status changes are saved and retrieved correctly.
RSK-03	Data loss or configuration errors affect the database or deployed application.	High	Low	Maintain database backups, document recovery procedures, and test backup restoration before final handover.
RSK-04	Lack of official access to institutional databases prevents live data synchronization.	Medium	High	Use authorized mock accounts and structured sample data. Do not attempt unauthorized access to institutional systems.

RSK-05	Development or integration tasks exceed the four-week project schedule.	High	Medium	Monitor weekly milestones, assign tasks according to department responsibilities, prioritize required features, and address delays early.
RSK-06	QR code verification fails or accepts an invalid clearance reference.	High	Medium	Test valid, invalid, and altered references and ensure verification is performed against the appropriate server-side records.
RSK-07	Automated email notifications fail because of service configuration or delivery problems.	Medium	Medium	Test the configured email service using test accounts, verify notification triggers, and document any service limitations.
RSK-08	Cloud deployment problems prevent the application from being accessed during the final demonstration.	High	Medium	Prepare the hosting environment early, test the deployed application before the final demonstration, and maintain a working local demonstration backup.
RSK-09	Project requirements or requested features change during development.	Medium	Medium	Record proposed changes in the project change log and obtain the required review before modifying the approved scope or schedule.

9. Project Management Plan and Document Updates

Control activity	Management approach
Review	Periodically review project documents, schedules, requirements, and deliverables against the approved project charter, the four-week schedule, and actual development progress.
Record Changes	Document proposed and approved changes to the project scope, system requirements, schedule, and deliverables in the project change log.
Version Control	Update the document version, revision history, and revision date whenever an approved change is incorporated into the Project Management Plan.
Communicate	Inform the Project Manager, department leads, team members, and relevant stakeholders of approved changes that affect their responsibilities or project deliverables.

10. Assumptions and Decisions Requiring Confirmation

Control activity	Management approach
Identify	Document foundational assumptions regarding system scope, technology stack, and third-party integrations.
Verify	Cross-check operational and technical assumptions with the project instructor, team lead, and institutional guidelines.
Track	Monitor pending confirmations to ensure they do not block critical development or deployment phases.
Record	Maintain a record of finalized project decisions to establish clear boundaries for system implementation.

10.1 Assumptions and Pending Decisions

11. Approval

Role / Stakeholder	Responsibility	Sign-off status
Project Manager	Review and endorse the overall Project Management Plan, scope, and deliverables.	
Project Instructor	Review and grant final formal approval of the project plan, requirements, and system setup	